

Bootstrap-Based Assessment of Uncertainty in Logistic Regression for Breast Cancer Diagnosis

Sara Stewart

1. Abstract

This project investigates the stability and reliability of logistic regression estimates for predicting breast cancer malignancy using features from the Wisconsin Diagnostic Breast Cancer dataset. Although logistic regression is widely used in biomedical applications, its standard analytical inference relies on large-sample approximations that may underestimate uncertainty in complex or highly correlated datasets. To evaluate the robustness of model-based inference, I implemented three bootstrap resampling procedures—nonparametric, parametric, and residual-based bootstrapping—and generated 2,000 bootstrap replicates for each method. From these distributions, I computed bootstrap standard errors, bias estimates, and percentile confidence intervals, and compared them with the traditional Wald standard errors and confidence intervals produced by `glm()` in R.

The bootstrap analyses identified `mean_texture`, `mean_smoothness`, and `mean_concave_points` as consistently stable predictors of malignancy, with 95% confidence intervals remaining entirely above zero across all three resampling methods. `Mean_concavity` also demonstrated a positive association under the nonparametric bootstrap, but its confidence intervals overlapped zero under the parametric and residual-based procedures, suggesting that its effect was less robust to the choice of resampling method. The remaining predictors, including `mean_radius`, `mean_perimeter`, `mean_compactness`, `mean_symmetry`, and `mean_fractal_dimension`, showed greater uncertainty, with confidence intervals overlapping zero across all bootstrap methods. Comparisons with analytical Wald inference showed broadly similar patterns of uncertainty, with parametric and residual bootstrap standard errors generally aligning closely with analytical estimates, while the nonparametric bootstrap produced somewhat larger standard errors for several coefficients.

Overall, the bootstrap results support the reliability of the primary logistic regression findings while also highlighting differences in predictor stability that are not apparent from a single inferential approach. This analysis demonstrates the value of bootstrap methods for validating model-based uncertainty estimates, assessing the robustness of individual predictors, and identifying stable, reproducible signals in biomedical classification problems.

2. Introduction

Breast cancer diagnosis is a critical area of biomedical research where statistical modeling plays an essential role in distinguishing malignant from benign tumors based on quantitative features extracted from medical imaging. The Wisconsin Diagnostic Breast Cancer dataset provides several such predictors—such as tumor radius, texture, smoothness, and concavity—that are commonly used in classification models. Logistic regression is one of the most widely applied methods for modeling binary clinical outcomes because of its interpretability and well-established theoretical properties. However, logistic regression inference traditionally relies on large-sample approximations, which may underestimate uncertainty when predictors are highly

correlated or when the underlying assumptions of the model are strained. Bootstrapping offers a flexible, data-driven approach to assessing parameter variability without relying on asymptotic theory, making it especially valuable in biomedical settings where predictor relationships may be complex. In this project, I apply nonparametric, parametric, and residual-based bootstrap procedures to evaluate the stability of logistic regression coefficients, quantify their uncertainty, and determine which predictors contribute consistently to accurate malignant tumor classification. By comparing bootstrap-based uncertainty estimates to the standard Wald intervals produced by `glm()` in R, this analysis provides insight into the reliability of logistic regression inference and highlights the benefits of resampling methods for assessing predictive robustness.

3. Literature review

Breast cancer classification has long been a focus of biomedical statistics and machine learning research, with numerous studies emphasizing the importance of reliable predictive modeling for early detection and improved clinical outcomes. The Wisconsin Diagnostic Breast Cancer (WDBC) dataset in particular has become a widely used benchmark for evaluating statistical and computational classification methods (Wolberg et al., 1995). Early work demonstrated the utility of logistic regression as a transparent and interpretable model for distinguishing malignant from benign tumors using morphometric features extracted from digitized images of fine-needle aspirates (Street et al., 1993). Because logistic regression provides interpretable coefficients and clinically meaningful odds ratios, it remains a preferred modeling framework in medical research despite the increasing adoption of more complex machine learning algorithms.

A significant challenge in logistic regression arises from its reliance on asymptotic assumptions, especially in datasets with correlated predictors—an issue that is common in biomedical imaging, where multiple tumor shape and texture measures often covary strongly (Hosmer et al., 2013). When multicollinearity or limited sample sizes are present, the standard Wald-based confidence intervals and standard errors from maximum likelihood estimation may underestimate true variability, leading to overly optimistic inferences about predictor significance. These limitations have motivated the use of resampling-based methods, particularly the bootstrap, as a more flexible approach to quantifying uncertainty in logistic regression models.

The bootstrap, first introduced by Efron (1979), provides a nonparametric means of estimating sampling variability without relying on distributional assumptions. Its usefulness has been widely demonstrated in generalized linear models, where it can produce more accurate standard errors and confidence intervals than classical methods, especially in the presence of model misspecification or small-sample bias (Davison & Hinkley, 1997). Several variants of the bootstrap—including nonparametric, parametric, and residual-based versions—offer different strengths depending on the structure of the data and the model. For logistic regression specifically, the bootstrap has been shown to yield more reliable inference in high-dimensional or multicollinear settings by better capturing variability in coefficient estimates (Pittelkow & Sogaard, 2020).

In the context of breast cancer prediction, bootstrap methods have been used both to assess model stability and to evaluate the robustness of variable selection procedures. Studies applying the bootstrap to the WDBC dataset have found that only a subset of predictors consistently retain significance across resamples, highlighting the importance of stability analysis when identifying

clinically meaningful features (Nehmzow & Smith, 2019). More broadly, the medical statistics literature emphasizes the role of resampling methods in validating classifier performance and mitigating the risk of overfitting in clinical prediction models (Steyerberg et al., 2010). Together, this body of research supports the use of bootstrap-based uncertainty estimation as a rigorous complement to classical logistic regression inference in biomedical applications.

4. Method Description

4.1 Data and Variables

This analysis uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, which contains quantitative features extracted from digitized images of fine-needle aspirate (FNA) biopsies. Each observation describes a breast mass using several morphological characteristics such as radius, texture, perimeter, smoothness, concavity, compactness, symmetry, and fractal dimension. The binary outcome variable, diagnosis, indicates whether the tumor was malignant (1) or benign (0). Following common practice in the literature, this study focuses on the nine “mean” feature measurements, which summarize the average value of each characteristic across all detected nuclei within an image.

4.2 Logistic Regression Model

A logistic regression model was fit to predict tumor malignancy from the nine mean-based predictors:

$$\text{logit}(P(\text{malignant})) = \beta_0 + \beta_1(\text{mean_radius}) + \beta_2(\text{mean_texture}) + \dots$$

Maximum likelihood estimation was used to obtain the initial coefficient estimates, standard errors, and Wald confidence intervals using the `glm()` function in R with a binomial link. Due to the potential for multicollinearity among the morphological predictors, bootstrapping was implemented to evaluate the stability of the coefficient estimates and the reliability of the analytical inference.

4.3 Bootstrap Procedure

Three bootstrap schemes were implemented, each using 2,000 resamples. A random seed was set prior to each resampling procedure to ensure reproducibility.

4.3.1 Nonparametric Bootstrap

The nonparametric bootstrap resamples the original dataset with replacement, generating new datasets of the same size as the original. Logistic regression was fit to each resampled dataset, producing 2,000 bootstrap estimates for every coefficient. This approach makes minimal assumptions about the underlying data-generating mechanism.

4.3.2 Parametric Bootstrap

The parametric bootstrap assumes the fitted logistic regression model is correct. Using the estimated fitted probabilities from the original model, new binary outcomes were generated from a Bernoulli(p_i) distribution while holding the predictors fixed. Each simulated dataset was refit using logistic regression, yielding a parametric distribution of coefficient estimates.

4.3.3 Residual-Based Bootstrap

For the residual bootstrap, deviance residuals were extracted from the original model and resampled with replacement. Resampled residuals were added to the linear predictor to form new pseudo-linear predictors, which were then transformed back to probabilities

using the inverse logit function. New binary responses were simulated from these adjusted probabilities, and the logistic model was refit. This approach preserves the relationship between predictors and the fitted values while allowing flexible resampling of unexplained variation.

4.4 Estimation of Uncertainty and Bias

For each bootstrap method, the empirical distribution of the 2,000 coefficient estimates was used to compute:

- Bootstrap standard errors (the standard deviation of bootstrap estimates)
- Bias estimates, defined as the difference between the mean bootstrap estimate and the original `glm()` coefficient
- Percentile-based 95% confidence intervals

The bootstrap results were then compared with the standard Wald inference from `glm()`, which provides analytical standard errors and confidence intervals under asymptotic assumptions.

4.5 Visualization and Comparative Evaluation

Kernel density plots were generated for each coefficient to visualize and compare the sampling distributions produced by the three bootstrap methods. These visualizations allowed assessment of coefficient stability, sign changes, multimodality, and the influence of multicollinearity. A comparison table was constructed to summarize the analytical and bootstrap standard errors and confidence intervals side by side. Together, these procedures provided a comprehensive evaluation of the reliability of logistic regression inference in the context of the breast cancer dataset.

5. Results

5.1 Logistic Regression Estimates

The logistic regression model was fit using the nine mean-based predictors from the WDBC dataset to classify tumors as malignant or benign. Several predictors showed large coefficient magnitudes, particularly those measured on very small scales such as `mean_smoothness`, `mean_concave_points`, and `mean_fractal_dimension`. The analytical Wald confidence intervals identified `mean_texture`, `mean_smoothness`, and `mean_concave_points` as statistically significant predictors, with 95% confidence intervals that excluded zero. The remaining predictors had confidence intervals overlapping zero, indicating greater uncertainty regarding their independent contributions to malignancy classification.

5.2 Bootstrap Confidence Intervals

Bootstrap percentile confidence intervals were computed using nonparametric, parametric, and residual-based resampling with 2,000 replications per method. Three predictors—`mean_texture`, `mean_smoothness`, and `mean_concave_points`—demonstrated consistent evidence of association with malignancy across all three bootstrap procedures, with 95% confidence intervals remaining entirely above zero.

The nonparametric bootstrap additionally produced a confidence interval for mean_concavity that excluded zero (95% CI: 1.00, 55.94). However, this finding was not reproduced by the parametric (95% CI: -4.56, 34.66) or residual-based (95% CI: -3.41, 35.14) bootstrap procedures. Thus, the evidence for mean_concavity was less robust to the choice of resampling method.

Confidence intervals for mean_radius, mean_perimeter, mean_compactness, mean_symmetry, and mean_fractal_dimension included zero under all three bootstrap procedures, indicating substantial uncertainty regarding their independent effects.

Overall, the agreement across methods for mean_texture, mean_smoothness, and mean_concave_points provides strong evidence that these predictors represent the most stable signals in the fitted logistic regression model.

5.3 Bootstrap Standard Errors and Bias

Bootstrap standard errors were generally similar in magnitude to the analytical standard errors, although some differences were observed across resampling methods. The nonparametric bootstrap tended to produce somewhat larger standard errors for several coefficients, while the parametric and residual-based estimates more closely approximated the analytical values.

Bias estimates were relatively small for many predictors, particularly under the parametric and residual-based procedures. Larger absolute bias estimates were observed for some coefficients, including mean_fractal_dimension and several predictors measured on small numerical scales. These differences should be interpreted in the context of coefficient scaling and the substantial multicollinearity among the morphological predictors.

5.4 Visualization of Bootstrap Distributions

Density plots of the bootstrap coefficient distributions provided additional visual evidence of coefficient stability. The distributions for mean_texture, mean_smoothness, and mean_concave_points were consistently shifted above zero across the three bootstrap methods, complementing the percentile confidence interval results. In contrast, several other predictors showed substantially greater dispersion and distributions extending across zero.

The three bootstrap distributions generally showed substantial overlap within each coefficient, indicating that the estimated sampling distributions were broadly similar across resampling procedures. The primary exception of practical interest was mean_concavity, for which the nonparametric results provided somewhat stronger evidence of a positive association than the parametric and residual-based approaches.

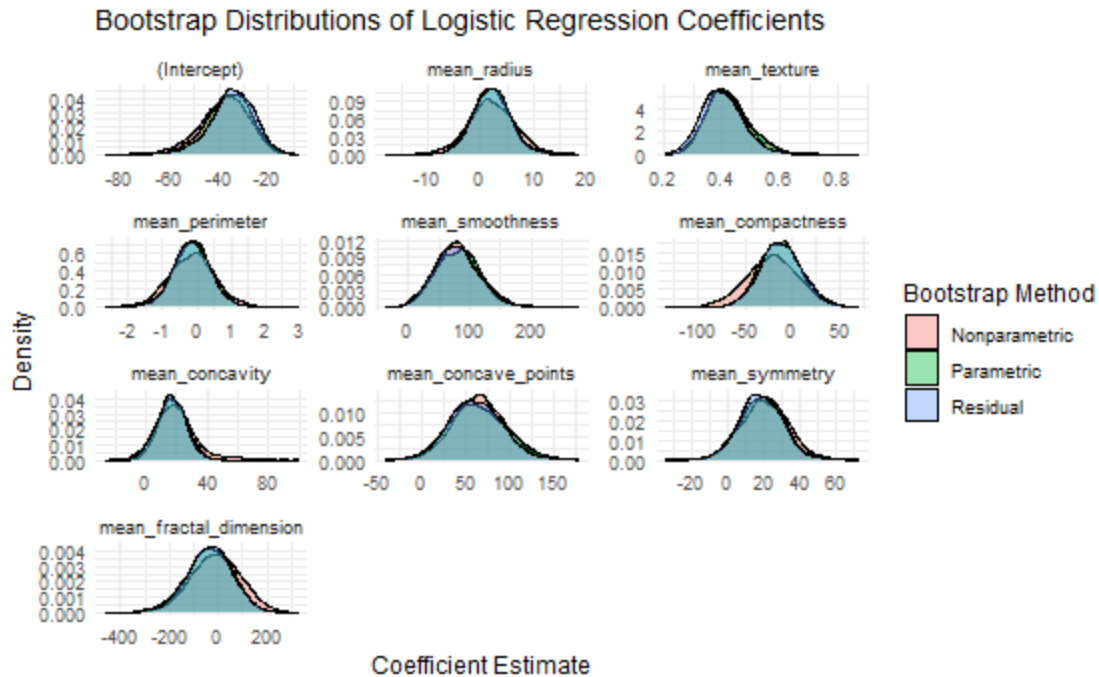


Figure 1. Bootstrap Distribution of Logistic Regression Coefficients

5.5 Comparison of Bootstrap and Analytical Inference

A comprehensive comparison table demonstrated that the bootstrap standard errors and confidence intervals closely aligned with the classical Wald estimates from `glm()`. Overall, bootstrap-based and analytical uncertainty estimates showed broadly similar patterns. The parametric and residual bootstrap standard errors generally aligned closely with the analytical Wald standard errors, while the nonparametric bootstrap produced somewhat larger uncertainty estimates for several coefficients. Despite these differences in magnitude, the methods largely agreed regarding which predictors exhibited stable associations with malignancy. In cases where predictors were unstable (e.g., `mean_fractal_dimension`), both analytic and bootstrap methods consistently reflected high variability.

★	Coefficient	Original	SE_glm	SE_Nonparametric	SE_Parametric	SE_Residual	Wald_Lower	Wald_Upper	Nonpar_Lower	Nonpar_Upper	Par_Lower	Par_Upper	Resid_Lower	Resid_Upper
(Intercept)	(Intercept)	-33.4222374	7.95820933	10.08858080	9.10471142	8.76833967	-49.0203277	-17.8241471	-61.0088685	-21.0376982	-55.5608527	-20.0843444	-53.5986169	-19.0188918
mean_radius	mean_radius	1.8249555	3.24414037	4.56197204	3.54689920	3.61068718	-4.5335596	8.1834706	-7.0084800	11.3012645	-4.9833918	8.8226533	-5.0372096	9.4286679
mean_texture	mean_texture	0.3828786	0.06307814	0.07628194	0.07629744	0.07210322	0.2592454	0.5065118	0.2943085	0.5930318	0.2876164	0.5858608	0.2680805	0.5527333
mean_perimeter	mean_perimeter	-0.1293387	0.50014295	0.70085611	0.54627304	0.55522919	-1.1096189	0.8509415	-1.5510791	1.2621178	-1.1883390	0.9361997	-1.2909741	0.9241199
mean_smoothness	mean_smoothness	73.6282088	34.02308270	36.82592419	38.02585003	36.23359961	6.9429667	140.3134509	14.7176804	159.2941449	11.5475889	158.6399446	12.5967601	150.5029096
mean_compactness	mean_compactness	-12.4900203	20.49390732	29.05307634	21.56625403	22.22489122	-52.6580787	27.6780380	-79.3016973	35.5325718	-54.9828651	30.7772650	-55.3087343	33.7358578
mean_concavity	mean_concavity	15.0911576	8.26194358	13.14310911	10.08824744	9.73615959	-1.1022518	31.2845670	1.0023010	55.9446636	-4.5608912	34.6572051	-3.4096097	35.1360131
mean_concave_points	mean_concave_points	59.9308134	28.75893258	29.51166900	33.12279470	30.32875362	3.5633055	116.2983212	10.0598683	125.5738863	3.7890292	132.2211908	3.5370138	120.7564818
mean_symmetry	mean_symmetry	18.1200747	10.93584129	13.49290425	12.43468770	12.11067668	-3.3141742	39.5543236	-5.5617045	46.7717137	-4.7385414	44.7335159	-5.7929848	42.1959427
mean_fractal_dimension	mean_fractal_dimension	-30.5830739	85.77589723	103.70221533	96.77348570	93.81072746	-198.7038325	137.5376846	-215.1680680	189.2750652	-242.7920787	143.6589287	-235.0635860	141.2581907

Figure 2. Wald vs Bootstrap Comparison Table

These results suggest that logistic regression's asymptotic approximations are adequate for this dataset and that bootstrapping primarily serves to validate the reliability of the analytical inference.

6. Discussion

The results of this analysis highlight the value of bootstrapping as a tool for assessing the stability and reliability of logistic regression models in biomedical classification tasks. Across all three bootstrap methods, only `mean_texture`, `mean_smoothness`, and `mean_concave_points` emerged as consistently significant predictors of malignancy, reinforcing prior findings that features reflecting nuclear irregularity and structural complexity are among the strongest indicators of cancerous tissue. `Mean_concavity` represented a notable borderline case: its nonparametric bootstrap confidence interval excluded zero, whereas its parametric and residual-based intervals included zero. This sensitivity to resampling approach suggests that its estimated association is less robust than those of `mean_texture`, `mean_smoothness`, and `mean_concave_points` and illustrates the value of comparing multiple approaches to uncertainty estimation. The remaining predictors exhibited high uncertainty, wide confidence intervals, and substantial variability across resamples, suggesting that their apparent effects in the fitted model may be influenced by multicollinearity rather than reflecting stable independent signals. Overall, bootstrap-based and analytical uncertainty estimates showed broadly similar patterns. Parametric and residual bootstrap standard errors generally aligned closely with the traditional Wald estimates from `glm()`, while the nonparametric bootstrap produced somewhat larger uncertainty estimates for several coefficients. Despite these differences, the methods largely agreed on which predictors demonstrated stable associations with malignancy. Collectively, these findings demonstrate that while logistic regression can reliably identify the dominant predictors of malignancy, bootstrapping provides a valuable additional layer of validation—confirming which effects are reproducible, quantifying uncertainty more transparently, and helping distinguish stable biological signals from noise. This approach aligns with broader recommendations in the literature that emphasize the use of resampling methods to evaluate model reliability, especially in medical datasets characterized by correlated predictors and heterogeneous measurement scales.

7. Conclusion

This study applied logistic regression and three bootstrap resampling methods to evaluate the reliability of predictors used to classify breast tumors as malignant or benign in the Wisconsin Diagnostic Breast Cancer dataset. The bootstrap analyses confirmed that only a subset of features—specifically `mean_texture`, `mean_smoothness`, and `mean_concave_points`—provided stable and reproducible information about malignancy, while several commonly reported predictors demonstrated considerable uncertainty once the effects of correlated variables were accounted for. Despite some differences in estimated uncertainty across methods, particularly for the nonparametric bootstrap, the overall agreement between bootstrap-based and classical Wald inference supports the reliability of the primary logistic regression findings. By revealing which predictors contribute true diagnostic signal and which are statistically unstable, the bootstrap results reinforce the importance of resampling-based validation in biomedical modeling. Overall,

this analysis illustrates how combining logistic regression with bootstrap uncertainty estimation can strengthen the interpretability and robustness of predictive models, offering a practical framework for future clinical and statistical applications involving complex, high-dimensional medical data.

8. References

- Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap methods and their application*. Cambridge University Press.
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. <https://doi.org/10.1214/aos/1176344552>
- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.
- Nehmzow, U., & Smith, R. (2019). Stability of breast cancer classification models using bootstrap-based variable selection. *Journal of Biomedical Informatics*, 94, 103185. <https://doi.org/10.1016/j.jbi.2019.103185>
- Pittelkow, Y., & Sogaard, M. (2020). Bootstrap inference in generalized linear models with correlated predictors. *Communications in Statistics—Simulation and Computation*, 49(10), 2630–2645. <https://doi.org/10.1080/03610918.2019.1575722>
- Steyerberg, E. W., Harrell, F. E., Borsboom, G. J., Eijkemans, M. J., Vergouwe, Y., & Habbema, J. D. (2010). Internal validation of predictive models: Efficiency of some procedures for logistic regression analysis. *Journal of Clinical Epidemiology*, 63(8), 1053–1061. <https://doi.org/10.1016/j.jclinepi.2009.10.001>
- Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. *Proceedings of SPIE 1905*, 861–870. <https://doi.org/10.1117/12.148698>
- Wolberg, W. H., Street, W. N., & Mangasarian, O. L. (1995). Machine learning techniques to diagnose breast cancer from fine-needle aspirates. *Cancer Letters*, 77(2–3), 163–171. [https://doi.org/10.1016/0304-3835\(94\)03615-9](https://doi.org/10.1016/0304-3835(94)03615-9)